

Hello, I'm currently following a paper on the Color Glass Condensate. I'm trying to understand how the classical treatment of the saturated gluon fields is justified. First, they imply that the unintegrated gluon density is of the order of $1/\alpha_s$, but I'm not sure where this estimate comes from. How is it justified?

You're diving into a fascinating topic! The Color Glass Condensate (CGC) framework aims to describe the high-energy limit of Quantum Chromodynamics (QCD), where the gluon fields are dense and non-linear effects become important. The saturated gluon fields in this regime can be treated classically under certain approximations, and understanding how this treatment is justified is key to understanding CGC.

The estimate that the unintegrated gluon density is of the order of $\frac{1}{\alpha_s}$ comes from considerations of gluon saturation in high-energy QCD. Here's a rough outline of how this estimate is typically justified:

1. High-energy limit and gluon density:

In the high-energy limit, especially in deep inelastic scattering (DIS) at small Bjorken x , gluons become densely populated in phase space. The density of these gluons can be described by the unintegrated gluon distribution function, which quantifies the number of gluons in a given phase space cell.

The physical situation is that at high energies, there are so many gluons that their interactions become important, and the gluon density grows rapidly. However, this growth is limited by non-linear effects, which eventually lead to saturation (the CGC regime).

The unintegrated gluon density $\tilde{f}(x, k_T)$, where k_T is the transverse momentum, becomes large in the high-energy limit but eventually saturates as x (the Bjorken variable) becomes very small.

2. Estimate from the classical field theory:

The classical treatment of the gluon fields is justified in the regime where the field strength is large, and interactions become strong. In this limit, gluons are highly occupied in phase space, and quantum effects such as quantum coherence are not the dominant

contribution. This regime can be modeled by a classical field theory.

The key to understanding the $1/\alpha_s$ estimate comes from the fact that the unintegrated gluon distribution $\tilde{f}(x, k_T)$ grows with decreasing x , but when the gluon density becomes large enough (i.e., when the number of gluons in a given transverse area is large), quantum fluctuations lead to a non-linear feedback that prevents the density from growing indefinitely. The saturation scale $Q_s^2(x)$ marks the boundary of this nonlinear regime and defines the typical momentum scale of the system.

The gluon density in this saturated regime is typically of the order $\tilde{f}(x, k_T) \sim \frac{1}{\alpha_s}$, as this is the scale at which non-linear effects become relevant and prevent further growth of the gluon distribution. This is analogous to the classical treatment of fields in statistical mechanics, where, at large occupations, the classical approximation is justified by the large occupation number.

3. Understanding the $1/\alpha_s$ scaling:

The $1/\alpha_s$ estimate for the unintegrated gluon density comes from the fact that in the CGC framework, the field strength becomes large and the gluon occupation number per mode scales with $1/\alpha_s$, where α_s is the strong coupling constant.

In more detail:

- **Classical field description:** In the CGC framework, the gluon fields are treated as classical fields in a regime where the quantum fluctuations are large but can be approximated classically. The field strength and the corresponding gluon density are such that the classical approximation becomes valid when the density becomes large enough.
- **Relation to the saturation scale:** The saturation scale Q_s is the momentum scale where the gluon density reaches a critical value, and this critical value is typically estimated to be of the order $\frac{1}{\alpha_s}$ in the high-energy limit, due to the balance between the gluon density growth and non-linear interactions that lead to saturation.

4. Summary:

The estimate that the unintegrated gluon density is of order $\frac{1}{\alpha_s}$ comes from considering the growth of the gluon density in high-energy QCD and the fact that when the density becomes sufficiently large, quantum effects lead to saturation, which halts the growth. This saturation condition is tied to the scale Q_s^2 , and the occupation number per gluon mode in this regime scales with $1/\alpha_s$, which is why this estimate is often quoted.

In short, this estimate arises from a combination of high-energy behavior, gluon saturation, and the classical treatment of the fields when the occupation number is large. The $1/\alpha_s$ scaling is tied to the transition from a quantum to a classical regime, where the gluon density becomes sufficiently high to justify a classical description.



Thank you. And where does the estimate of the occupation numbers come from?

Great question — this is really the core of why the CGC can be treated classically.

The estimate of *large occupation numbers* comes from a combination of (i) how gluons are produced at weak coupling and (ii) the condition for saturation, i.e. when nonlinear gluon interactions become as important as linear emission.

Let me walk you through the logic in a way that is standard in the CGC literature.

1. What is meant by “occupation number” for gluons?

Roughly, the occupation number $f(k)$ is the number of gluons per unit phase-space cell,

$$f(k) \sim \frac{1}{(2\pi)^3} \frac{dN}{d^3k}.$$

In a gauge theory, this can be related to the gauge field amplitude:

$$f(k) \sim \frac{k}{g^2} \langle A_i^a(k) A_i^a(-k) \rangle.$$

So large occupation numbers correspond to large classical field amplitudes.

2. Weak coupling does *not* mean small fields

In the CGC regime:

- The coupling $g \ll 1$
- But the color charge density of fast partons is large

This is crucial. The Yang–Mills equations schematically read:

$$D_\mu F^{\mu\nu} = J^\nu$$

with $J^\nu \sim g\rho$, where ρ is the color charge density of fast (large- x) partons.

Because ρ is large, the classical solution satisfies

$$A \sim \frac{1}{g}.$$

This is *not* an assumption — it follows self-consistently from solving the equations with a dense source.

3. From field strength to occupation number

Now estimate the occupation number using dimensional analysis.

For modes with momentum $k \sim Q_s$:

- Field amplitude: $A \sim 1/g$
- Field strength: $F \sim kA \sim Q_s/g$

The energy density is

$$\varepsilon \sim F^2 \sim \frac{Q_s^2}{g^2}.$$

On the other hand, if we think of this as a gas of gluons with occupation number f ,

$$\varepsilon \sim \int d^3k k f(k) \sim Q_s^4 f(Q_s).$$

Equating the two:

$$Q_s^4 f(Q_s) \sim \frac{Q_s^4}{g^2} \Rightarrow f(Q_s) \sim \frac{1}{g^2} \sim \frac{1}{\alpha_s}.$$

This is the origin of the estimate.

4. Saturation as a balance of linear and nonlinear effects

There is also a very intuitive dynamical argument.

- Linear small- x evolution (BFKL) causes gluon densities to grow rapidly.
- Nonlinear interactions (gluon recombination) enter when:

$$\text{interaction rate} \sim \text{emission rate}.$$

Parametrically:

- Emission rate $\sim \alpha_s f$
- Recombination rate $\sim \alpha_s^2 f^2$

Saturation occurs when these are comparable:

$$\alpha_s f \sim \alpha_s^2 f^2 \Rightarrow f \sim \frac{1}{\alpha_s}.$$

Again, the same scaling appears.

5. Why large occupation $\Rightarrow$ classical fields

A bosonic field behaves classically when:

$$f(k) \gg 1.$$

In CGC:

$$f(k \lesssim Q_s) \sim \frac{1}{\alpha_s} \gg 1 \quad (\text{since } \alpha_s \ll 1).$$

Thus:

- Quantum corrections are suppressed by powers of $1/f \sim \alpha_s$
- The dominant dynamics is classical Yang–Mills evolution
- Quantum effects enter only through **slow evolution of the sources** (JIMWLK/BK)

This is exactly the “Color Glass” separation of scales.

6. Summary (key chain of reasoning)

$$\text{Dense sources} \Rightarrow A \sim \frac{1}{g} \Rightarrow f \sim \frac{1}{\alpha_s} \Rightarrow \text{classical gluon fields}$$

So the large occupation number is not assumed — it *emerges* from weak coupling combined with high gluon density and the saturation condition.

If you want, next we can:

- Derive this explicitly in the McLerran–Venugopalan model, or
- Connect it to the unintegrated gluon distribution $\phi(x, k_T)$, or
- Discuss how quantum corrections are organized around this classical background.



Thank you. Could you show me why in your section 1 the occupation number is related to a two-point function of the gluon fields?

Yes — this is an important point, and it’s good to see it explicitly rather than just stated. I’ll keep things as concrete and standard as possible, and I’ll be careful about what is schematic versus precise.

The short answer is: **for weakly coupled gauge fields, the gluon occupation number is extracted from the equal-time two-point function of the gauge field (or field strength), exactly as for any bosonic field.** The CGC just uses this in a regime where that two-point function is large.

Let me derive this step by step.

1. Warm-up: scalar field (cleanest derivation)

Start with a free scalar field (this avoids gauge subtleties and shows the logic clearly):

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_{\mathbf{k}} e^{-ik \cdot x} + a_{\mathbf{k}}^\dagger e^{ik \cdot x} \right), \quad \omega_k = |\mathbf{k}|.$$

The occupation number operator is

$$\hat{n}_{\mathbf{k}} = a_{\mathbf{k}}^\dagger a_{\mathbf{k}}.$$

Now compute the equal-time two-point function in momentum space:

$$\langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle = \frac{1}{2\omega_k} \left(\langle \hat{n}_{\mathbf{k}} \rangle + \frac{1}{2} \right).$$

Ignoring the vacuum $\frac{1}{2}$ (irrelevant when $f \gg 1$):

$$\langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle \sim \frac{f(k)}{\omega_k}.$$

Rearranging:

$$f(k) \sim \omega_k \langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle.$$

This is the key structural relation.

2. Vector field: gluons in Coulomb or light-cone gauge

Now move to gluons. Fix a physical gauge (e.g. Coulomb gauge or $A^+ = 0$), so only transverse polarizations propagate.

The mode expansion is schematically:

$$A_i^a(x) = \sum_{\lambda=1,2} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\epsilon_i^\lambda(\mathbf{k}) a_{\mathbf{k},\lambda}^a e^{-ik \cdot x} + \text{h.c.} \right].$$

The equal-time two-point function becomes:

$$\langle A_i^a(\mathbf{k}) A_j^b(-\mathbf{k}) \rangle = \delta^{ab} \left(\delta_{ij} - \frac{k_i k_j}{k^2} \right) \frac{1}{2\omega_k} \left(f(k) + \frac{1}{2} \right).$$

Again, dropping the vacuum piece:

$$f(k) \sim \omega_k \langle A_i^a(\mathbf{k}) A_i^a(-\mathbf{k}) \rangle.$$

This is the precise origin of the relation I wrote schematically earlier.

3. Why a factor of $1/g^2$ appears in CGC estimates

In the CGC literature you often see

$$f(k) \sim \frac{k}{g^2} \langle AA \rangle.$$

This is because the **classical field is typically written with a factor of $1/g$** :

$$A_\mu = \frac{1}{g} \mathcal{A}_\mu,$$

where $\mathcal{A}_\mu \sim \mathcal{O}(1)$.

Then:

$$\langle AA \rangle \sim \frac{1}{g^2} \langle \mathcal{A} \mathcal{A} \rangle,$$

and plugging into the previous relation gives:

$$f(k) \sim \frac{k}{g^2}.$$

This is not a new physical assumption — it is just bookkeeping for weak coupling.

4. Gauge-invariant version (often preferred)

To avoid gauge dependence, one can define the occupation number using the field

strength:

$$f(k) \sim \frac{1}{k^2} \langle F_{ij}^a(\mathbf{k}) F_{ij}^a(-\mathbf{k}) \rangle.$$

Since $F_{ij} \sim kA$, this is equivalent to the previous definition.

This is the form commonly used in lattice CGC simulations.

5. Why this definition makes sense in CGC

This definition relies on:

- Weak coupling $g \ll 1$
- High occupancy $f \gg 1$
- Modes being approximately on-shell ($k^0 \sim |\mathbf{k}|$)

All of these hold for modes with $k \lesssim Q_s$.

Thus, although the theory is interacting, **the notion of occupation number is well defined parametrically.**

6. Summary

The chain of logic is:

1. For any bosonic field:

$$f(k) \sim \omega_k \langle \phi(k) \phi(-k) \rangle$$

2. For gluons in a physical gauge:

$$f(k) \sim k \langle A_i^a(k) A_i^a(-k) \rangle$$

3. In CGC:

$$A \sim \frac{1}{g} \Rightarrow f(k) \sim \frac{1}{\alpha_s}$$

This is why the two-point function of classical gluon fields directly encodes the occupation number.

If you'd like, next I can:

- Show this explicitly in the McLerran–Venugopalan model, or
- Connect it to the unintegrated gluon distribution $\phi(x, k_T)$, or

- Discuss limitations of the occupation-number concept beyond $k \sim Q_s$.



Thank you. Could you show me explicitly how you computed the two-point function in section 1?

Of course — let's do it explicitly and carefully, with no hand-waving. I'll work through the scalar field example in full detail, because that is exactly the computation being implicitly used for gluons.

1. Setup: free scalar field

Consider a free, real scalar field in a box (or infinite volume; the result is the same):

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_{\mathbf{k}} e^{-i\omega_k t + i\mathbf{k} \cdot \mathbf{x}} + a_{\mathbf{k}}^\dagger e^{i\omega_k t - i\mathbf{k} \cdot \mathbf{x}} \right), \quad \omega_k = |\mathbf{k}|.$$

The creation and annihilation operators satisfy

$$[a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}').$$

2. Field in momentum space

Define the spatial Fourier transform at fixed time t :

$$\phi(\mathbf{k}, t) \equiv \int d^3x e^{-i\mathbf{k} \cdot \mathbf{x}} \phi(\mathbf{x}, t).$$

Insert the mode expansion:

$$\phi(\mathbf{k}, t) = \frac{1}{\sqrt{2\omega_k}} \left(a_{\mathbf{k}} e^{-i\omega_k t} + a_{-\mathbf{k}}^\dagger e^{i\omega_k t} \right).$$

3. Compute the two-point function

We now compute the **equal-time two-point function**:

$$\langle \phi(\mathbf{k}, t) \phi(\mathbf{k}', t) \rangle.$$

Insert the expressions:

$$\langle \phi(\mathbf{k}) \phi(\mathbf{k}') \rangle = \frac{1}{\sqrt{2\omega_k 2\omega_{k'}}} \left\langle (a_{\mathbf{k}} + a_{-\mathbf{k}}^\dagger) (a_{\mathbf{k}'} + a_{-\mathbf{k}'}^\dagger) \right\rangle.$$

Expand all terms:

$$\begin{aligned} \langle a_{\mathbf{k}} a_{\mathbf{k}'} \rangle &= 0, \\ \langle a_{-\mathbf{k}}^\dagger a_{-\mathbf{k}'}^\dagger \rangle &= 0, \\ \langle a_{-\mathbf{k}}^\dagger a_{\mathbf{k}'} \rangle &= (2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}') f(k), \\ \langle a_{\mathbf{k}} a_{-\mathbf{k}'}^\dagger \rangle &= (2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}') [1 + f(k)]. \end{aligned}$$

Here

$$f(k) \equiv \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle$$

is the occupation number.

4. Final result

Putting everything together:

$$\begin{aligned} \langle \phi(\mathbf{k}) \phi(\mathbf{k}') \rangle &= \frac{(2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}')}{2\omega_k} [f(k) + 1 + f(k)] \\ &= \frac{(2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}')}{2\omega_k} (2f(k) + 1). \end{aligned}$$

Often this is written as

$$\boxed{\langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle = \frac{1}{\omega_k} (f(k) + \frac{1}{2})}$$

The $\frac{1}{2}$ is the vacuum (zero-point) contribution.

5. Extracting the occupation number

When the occupancy is large ($f \gg 1$), as in CGC:

$$\langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle \approx \frac{f(k)}{\omega_k}.$$

Thus,

$$f(k) \approx \omega_k \langle \phi(\mathbf{k}) \phi(-\mathbf{k}) \rangle$$

This is the precise statement used in CGC.

6. Extension to gluons (why nothing essential changes)

For gluons in a physical gauge:

- Replace $\phi \rightarrow A_i^a$
- Sum over transverse polarizations
- Include color index a

The structure remains:

$$\langle A_i^a(\mathbf{k}) A_j^b(-\mathbf{k}) \rangle \propto \delta^{ab} \left(\delta_{ij} - \frac{k_i k_j}{k^2} \right) \frac{1}{\omega_k} \left(f(k) + \frac{1}{2} \right).$$

That is why the **equal-time two-point function directly measures the occupation number**.

7. Why this matters for CGC

Since in the CGC

$$\langle AA \rangle \sim \frac{1}{g^2},$$

this immediately implies

$$f(k) \sim \frac{1}{\alpha_s} \gg 1,$$

justifying:

- Classical Yang–Mills fields
- Suppression of quantum corrections
- The entire CGC framework

If you'd like, next we can:

- Repeat this derivation using **field strengths** $F_{\mu\nu}$ (fully gauge invariant), or
- Show how this appears explicitly in the **McLerran–Venugopalan model**, or
- Discuss when and why this occupation-number picture breaks down.

